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Expert Review of Molecular Diagnostics

ISSN: 1473-7159 (Print) 1744-8352 (Online) Journal homepage: www.tandfonline.com/journals/iero20

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To cite this article: Danyang Li, Joerg Patrick Burgstaller & Joanna Poulton (2025) How mitochondrial DNA complicates pre-implantation genetic screening and management, Expert Review of Molecular Diagnostics, 25:10, 637-640, DOI: 10.1080/14737159.2025.2545967

To link to this article: https://doi.org/10.1080/14737159.2025.2545967

Published online: 19 Aug 2025.

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EXPERT REVIEW OF MOLECULAR DIAGNOSTICS
2025, VOL. 25, NO. 10, 637–640
https://doi.org/10.1080/14737159.2025.2545967

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EDITORIAL

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How mitochondrial DNA complicates pre-implantation genetic screening and management

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ARTICLE HISTORY Received 28 May 2025; Accepted 6 August 2025

KEYWORDS Mitochondrial DNA; heteroplasmy; embryo; mtDNA inheritance; mitochondrial replacement therapy (MRT); pre-implantation genetic screening (PGS); pre-implantation genetic testing (PGT)

1. Introduction

Mitochondrial DNA (mtDNA) disorders, caused by maternally inherited point mutations, are relatively common; approximately 1 in 400 individuals carry such mutations, which can often lead to multisystem disease [1]. Heteroplasmy is a key characteristic of these disorders, meaning that patients may harbor both mutant and normal mtDNA. Clinical symptoms depend on exceeding specific threshold levels of mutant mtDNA. Pre-implantation genetic testing (PGT), otherwise known as pre-implantation genetic screening (PGS), is a technique used in assisted reproduction that aims to reduce the transmission of these disorders by screening embryos for pathogenic mtDNA mutations to assess disease risk. However, both variability in mutant load and technical challenges in assessing heteroplasmy complicate selection of pre-implantation embryos. This editorial reviews PGT methodologies for mtDNA disorders, their limitations, and emerging strategies to improve diagnostic accuracy and clinical outcomes, updating our previous international consensus [2]. These are summarized in Table 1.

2. The inheritance of mitochondrial DNA

Unlike nuclear DNA, mtDNA is exclusively maternally inherited. In most human somatic cells, mtDNA copies ranges from a few hundred to several thousand per cell, depending on tissue type, whereas each oocyte contains approximately 50,000 to 330,000 mtDNA copies [11]. The genetic bottleneck during oogenesis results in rapid intergenerational shifts in heteroplasmy levels [12], complicating predictions of disease risk in offspring. Segregation depends on factors such as maternal age, embryonic developmental stage, and mitochondrial haplotype [13]. The bottleneck size may be orders of magnitude lower than the total mtDNA content, potentially as few as one, which dramatically affects heteroplasmy segregation. For instance, different mutations such as m.8993T > G/C, segregate more rapidly than others like m.3243A > G [14]. After fertilization, mtDNA replication is paused and resumes later in development, mtDNA content being determined by partitioning among blastomeres during early cleavage stages [15]. In the absence of selection, heteroplasmy varies significantly among embryos, but the average heteroplasmy level across all embryos is close to that of the mother [9]. Although segregation continues during embryonic development, studies such as Sallevelt et al. have shown that in approximately 80% of embryos, heteroplasmy levels are relatively consistent across blastomeres [15]. This consistency suggests that analysis of a single blastomere is generally reliable for PGT of mtDNA mutations, though notable variation in a subset of embryos (approximately 11%) means that some risk of misdiagnosis persists [15].

3. Heteroplasmy and the threshold effect

Clinical symptoms of mitochondrial diseases typically manifest when the heteroplasmy level crosses a specific threshold, which varies depending on the mutation and tissue type. For example, the threshold for the m.8993T > G mutation, (linked to Neurogenic Muscle Weakness, Ataxia, and Retinitis Pigmentosa, or NARP), to cause clinical symptoms is a mutant load of over 90%, whereas the m.3243A > G mutation (linked to Mitochondrial Encephalomyopathy, Lactic Acidosis, and Stroke-like episodes, or MELAS) can manifest at lower heteroplasmy levels (around 50–60% in muscle) [16]. Hellebrekers et al. found that muscle mutation levels of the m.3243A > G mutation below 18% were associated with a 95% probability of being asymptomatic [17], but the dynamics of mutant load are complex, so blood levels usually underestimate systemic mutation load. For example, m.3243G mutant mtDNA accumulates in post-mitotic tissues but tend to decrease with age in blood [18]. Moreover, Sacconi et al. reported an mtDNA anticodon mutation, m.5545C > T in the $tRNA^{Trp}$ gene that behaves as a functional dominant [19]. The threshold was < 25% for a severe phenotype in affected tissues but was only 4–8% *ex vivo*. Furthermore, thresholds depend not only on load of specific mutation and tissue type but also on genetic background and environmental stressors.

4. Technical challenges in PGT for mtDNA disorders

PGT is a preventative technique used in assisted reproduction to detect abnormalities in the genetic material of embryos.

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Table 1. Comparison of genetic testing methods at different developmental stages.

Biopsy Stage	Advantages	Limitations	References
Polar body biopsy	Noninvasive, avoids harm to developing embryo.	Limited predictive accuracy for embryo heteroplasmy, mismatch between polar body and oocyte mtDNA levels	[3]
Cleavage-stage biopsy (Day 3)	Provides early genetic information, allowing timely embryo transfer.	Higher mosaicism, potential harm to embryo development	[4,5]
Blastocyst (TE) biopsy (Day 5–6)	Less detrimental to embryo, uncertain accuracy of heteroplasmy level.	Limited data on correlation with ICM, heteroplasmy differences up to 23% observed in some blastocysts	[6–8]
Amniocentesis (post-implantation, ~15–20 weeks)	Definitive diagnosis using fetal cells; high accuracy for mtDNA heteroplasmy.	Performed post-implantation; time constraints if pregnancy termination is considered.	[9,10]

Abbreviations: mtDNA – mitochondrial DNA; PGT – Pre-implantation Genetic Testing; MRT – Mitochondrial Replacement Therapy; NGS – Next Generation Sequencing; TE – Trophectoderm; ICM – Inner Cell Mass.

While the alternative of oocyte donation completely eliminates the risk, PGT is currently the most widely used reproductive strategy to reduce the transmission of mtDNA disorders. Ideally, the goal is to transfer embryos with no detectable mutant mtDNA. However, when this is not possible, embryos with mutation levels below a certain threshold may be considered for transfer. The challenge lies in defining this 'safe' threshold, as it varies depending on the specific mtDNA mutation and its correlation with disease severity. For instance, thresholds of 15% for the m.3243A > G mutation and 30% for the m.8993T > G mutation have been proposed to minimize the risk of severe disease manifestation [4]. For most mutations, health risks could be substantially reduced by setting a low cutoff point. However, the lower the cutoff points are fixed, the more embryos are discarded, and thus, the lower the 'take home baby rate.' On the other hand, the higher the cutoff points are fixed, the more embryos will probably be available, but with an increased risk that the welfare of the child will be compromised. These proposed cutoff points should be regarded as provisional and adjusted as more experience with PGT for mtDNA mutations is gained.

Moreover, heteroplasmy levels measured during biopsy are not fixed; they can change post-implantation due to tissue-specific segregation, and other factors. These post-implantation dynamics complicate the prediction of long-term outcomes for the child. Blastomeres and trophectoderm (TE) cells may show significant differences in mutation load, which could lead to discrepancies in diagnosis. In some cases, interblastomere variation of up to 24% has been observed, which could result in false positives or false negatives if only a single blastomere is sampled [4,6]. Although Treff et al. demonstrated a strong correlation between mutation load in TE and the inner cell mass, the findings remain controversial because the mutant load harbored by an infant from the 'low risk' embryo (12% m.3243A > G in TE) [20] appeared to have risen to 42–52% [7]. Sampling multiple cells may mitigate diagnostic error but affects invasiveness and impact of the biopsy on embryo viability [5]. This represents a key challenge in setting a safe mutation threshold (Table 1).

Therefore, embryo biopsy at different developmental stages plays a crucial role in PGT for mitochondrial disorders, each with its own advantages and limitations. Biopsy can be performed at the cleavage stage (day 3), blastocyst stage (day 5–6), or at the (Metaphase II) MII oocyte stage and/or the zygote stage by analyzing the first and/or second polar body (Table 1). Studies have shown that TE biopsies at blastocyst generally align well with the mutation levels in the whole embryo [6]. Blastocyst biopsy is potentially more invasive but more accurate [15]. Polar body biopsy is even less invasive, but less accurate. TE biopsy is increasingly preferred for convenience, though more validation studies are needed. In addition, amniocentesis later in pregnancy can provide a more representative assessment of fetal heteroplasmy levels than chorionic villus sampling (CVS), because it samples fetal not trophoblast cells. However, by being performed later in gestation it introduces time constraints on decision-making [10].

5. New strategies and future directions to improve PGT in mitochondrial diseases

Recent strategies to improve PGT in mitochondrial diseases center on enhancing diagnostic accuracy while minimizing embryo harm. Advanced genomic methods, such as next-generation sequencing (NGS), now allow for high-resolution quantification of heteroplasmy and more reliable determination of mutation load in mtDNA, which is critical for selecting embryos with a healthy genetic profile. In parallel, non-invasive testing approaches – such as the analysis of cell-free DNA from blastocoel fluid or spent culture medium – are being optimized to avoid the potential risks of invasive biopsies while still providing detailed genetic insights [21].

6. Regulatory and ethical considerations for PGT in mitochondrial disorders

PGT for nuclear gene disorders is well-established worldwide, but PGT for mitochondrial disorders faces stricter regulations due to ethical concerns and uncertainties regarding long-term outcomes. National PGT regulations vary. In the UK, the Human Fertilisation and Embryology Authority (HFEA) permits PGT for specific mtDNA mutations. the U.S. allows PGT practice is variable because there are no federal regulations. Others, including Germany and France, impose strict limitations due to ethical concerns over embryo selection and potential germline modifications [22].

Some argue that selecting embryos based on heteroplasmy levels is ethically acceptable if it reduces the risk of severe disease. However, concerns arise because viable embryos are discarded due to uncertain thresholds. Additionally, mitochondrial replacement therapy (MRT), which involves replacing mutated mtDNA with donor mtDNA, is permitted in the UK but remains banned in many countries which hold that it is a genetic

modification. While MRT techniques like maternal spindle transfer have shown promise in overcoming developmental arrest caused by ooplasmic defects in animal models, resulting in healthy offspring with minimal mtDNA carryover, success is not guaranteed. Outcomes may vary depending on the technique, mutation type, and cellular context, the recipient mother's mtDNA rising to 60% in 1/6 pregnancies [23]. Reassuringly, a recent UK study [24] showed that MRT using pronuclear transfer resulted in eight live births, with newborn heteroplasmy reduced by up to 100%, though three had residual loads of 5%, 12% and 16%. The investigators compared MRT with PGT which led to 18 live births, with heteroplasmy up to 7%. Both approaches lowered transmission risk, but neither ensured complete elimination of mutant mtDNA. MRT had lower fertilization and embryo viability rates, less complete mutant clearance and higher biochemical pregnancy loss. It is best suited to women with homoplasmic [2] or high heteroplasmy mutations where PGT is less likely to succeed.

Given the complexity of mitochondrial inheritance, genetic counseling is paramount for families considering PGT or MRT for mitochondrial disorders. Prospective parents must be made aware of their limitations. This includes the possibility of false-negative results and of embryos that meet current mutation thresholds still exhibiting disease symptoms. While encouraging studies suggest a good outcome in early follow-up of children born following PGT, others suggest that assisted reproductive technology (ART) may generate some negative sequelae [25]. Larger cohort studies with long-term monitoring are necessary to confirm these findings. Ongoing dialogue among all stakeholders will be essential to ensure that these advanced reproductive technologies are applied responsibly and with the full consent and understanding of the families involved.

7. Conclusion

PGT offers a crucial tool for reducing the transmission of mitochondrial DNA disorders, but it is complicated by heteroplasmy dynamics, technical limitations, and ethical concerns. Advances in sequencing, noninvasive testing, and potential therapeutic strategies may enhance its accuracy and effectiveness. We are optimistic that refining selection thresholds, standardizing protocols, and ensuring long-term follow-up will improve clinical outcomes and support informed reproductive choices.

Funding

J Poulton and D Li were supported by the John Fell Fund (University of Oxford) [0016010] and the Charterhouse European Bursary (University of Oxford) [-]. Historic collaborative efforts were supported by the European Neuromuscular Centre workshop (2018). The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Declaration of interest

The authors have no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties.

Reviewer disclosures

Peer reviewers on this manuscript have no relevant financial or other relationships to disclose.

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